

The effects of carbon nanotubes, blend composition and glycidyl methacrylate-grafted polypropylene compatibilizer on the morphology, mechanical and electrical properties of polypropylene-polyamide 6 blends

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Abstract The effect of multi-walled carbon nanotubes (MWCNTs) and a glycidyl methacrylate grafted polypropylene (PP-g-GMA) on the morphology, electrical and mechanical properties of polypropylene (PP), polyamide 6 (PA6) and their respective blends (PP/PA6), prepared by melt mixing, was investigated. The study of MWCNT nanocomposites from PP/PA6 blends is of significant interest, given that they combine the thermomechanical properties of PA6 with the processability of PP, resulting in materials with a wide range of mechanical and electrical properties. In this work, it was demonstrated that the addition of the same amount of MWCNTs to immiscible PP/PA6 blends produced nanocomposites with different morphologies, mechanical and electrical properties depending on the polymer proportion, and that the PP-g-GMA improved mechanical properties while preserving the electrical properties. SEM and TEM images showed that MWCNTs were comparatively better dispersed in PA6 than in PP; however, the addition of PP-g-GMA to PP matrix improved MWCNT dispersion. On the other hand, SEM micrographs showed different morphologies of PP/PA6 blends, depending on the composition of the polymers in the blend. The addition of PP-g-GMA to PP/PA6 blends and PP/PA6/MWCNT nanocomposites reduced significantly the domain size in both dispersed and continuous phases. Regardless of compatibilization, MWCNTs were located preferentially in the PA6 phase. The electrical characterization showed that PP/PA6/MWCNT nanocomposites exhibited lower volume resistivity values ($10^8 \Omega \text{ cm}$) compared to PA6/MWCNT nanocomposites ($10^{11} \Omega \text{ cm}$), at the carbon nanotube load employed, as a result of the confinement of the MWCNTs in the PA6 phase. The incorporation of PP-g-GMA to the PP/PA6 blends and PP/PA6/MWCNT nanocomposites did not cause a significant change in

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the volume resistivity of the blends; moreover, PP-g-GMA considerably improved the modulus and tensile strength of the PP/PA blends (i.e. up to 90 and 97 %, respectively for PP/PA 80/20 blend).

Introduction

Since their discovery in 1991, carbon nanotubes (CNT) have been widely used in the development of polymer conductive materials due to their high aspect ratio and excellent electrical and mechanical properties [1–3]. A crucial factor to be considered in the design of conductive polymeric material is the conductive filler content. The electrical resistivity of the conductive composites decreases with conductive filler content; however, the mechanical properties are diminished at high filler content [4, 5]. Therefore, it is desirable to use the lowest content of conductive filler in order to achieve sufficient electrical properties for a particular application and at the same time to preserve the physical and mechanical properties of composites. Several publications have reported different approaches to synthesize polymer/nano-carbon composites with improved electrical and mechanical properties [6–14]. An effective strategy for reducing the content of conductive filler and thus improve the electrical conductivity is to use immiscible polymer blends. In these systems, the conductive filler is selectively located at the interface or within one polymer phase. As a result, the electrical conductivity is improved at low concentrations of conductive filler; moreover, the conductivity is usually higher in comparison with the corresponding composites from the individual polymer of the blend [15], which can be attributed to the double percolation phenomenon [16]. The production of conductive MWCNT nanocomposites from immiscible polymer blends has received considerable interest from the scientific community since it is possible to obtain materials with improved electrical properties at low MWCNT content [5, 15–25].

The immiscible polypropylene/polyamide 6 (PP/PA6) blends have significant commercial interest because they combine the properties of their components [26]. PA6 is an engineering polymer with excellent mechanical, thermal and solvent resistance properties; however, its MWCNT composites present low electrical conductivity, even at high MWCNT loads [15]. On the other hand, PP is a commodity resin showing good moisture resistance and processability and its MWCNT nanocomposites exhibit relatively good conductivity at lower filler content [27]. Thus, the electrical properties of PP/PA6/MWCNT composites are superior in comparison with those of PA6/MWCNTs [15].

The mechanical properties of immiscible polymer blends depend on the morphology and interface quality. In most cases, PP/PA6 systems show poor adhesion and high interfacial tension, exhibiting poor mechanical properties [28, 29]. One way to improve the interfacial adhesion is by using compatibilizing agents [29–32]. Traditionally, PP/PA6 blends have been compatibilized mainly through the use of polypropylene grafted maleic anhydride (PP-g-MA) [30, 33–35]. However, polyolefins grafted with glycidyl methacrylate (GMA) could be employed as compatibilizer; these materials present higher reactivity towards the amine,

hydroxyl and carbonyl functional groups in order to form covalent bonds across the interfaces due to the epoxide groups [36]. Additionally, compatibilizers containing GMA do not produce water molecules during the reaction with either $-\text{COOH}$ or $-\text{NH}_2$ groups from polyamides [37]. Very few studies have reported the use of PP-g-GMA as compatibilizer in immiscible PP/PA6 blends [38, 39].

Thus, compatibilized PP/PA6/MWCNT nanocomposites may exhibit higher electrical and mechanical properties than PA6/MWCNT nanocomposites. A few investigations have been published on the topic of PP/PA6/MWCNT composites, focusing on their morphology or electrical properties [15, 20, 33]; however, the combined effect of MWCNTs and a compatibilizing agent on the mechanical and electrical properties has not been studied. The aim of this work was to study the effect of MWCNTs and a compatibilizer (PP-g-GMA) on the morphology, electrical and mechanical properties of PP, PA6 and their blends, at different blend compositions.

Experimental

Materials

MWCNTs were purchased from Cheaptubes Inc. According to the manufacturer, the MWCNTs consist of MWCNT agglomerates with an outer mean diameter of 30–50 nm, inner mean diameter in the range of 5–10 nm and specific surface area of 60 m²/g. Polyamide-6 (PA6, TebeNyl 3100) (MFR 23 ± 5 g/10 min, M_w = 39,604 g/mol and density 1.12 g/cm³) was purchased from Thermoplastics del Caribe SA de CV (México). An extrusion grade polypropylene (MFR 3.8 g/10 min, M_w = 449,536 g/mol and density 0.9 g/cm³) was obtained from Indelpro Company. PP-g-GMA Resalloy 155 (GMA 5 wt%, MFR 4 g/10 min, M_w = 183,186 g/mol and density 1.05 g/cm³), kindly provided by Resirene SA de CV, Mexico, was used as compatibilizer.

Sample preparation

Immiscible PP/PA6 blends were prepared at different polymer proportions in order to study the effect of blend composition on the morphology and properties of composites, the corresponding uncompatibilized PP/PA6/MWCNT nanocomposites were prepared using a constant MWCNT content (4 wt%). Similarly, both compatibilized PP/PA6 blends and PP/PA6/MWCNT nanocomposites were prepared adding PP-g-GMA to the PP phase (5 wt%). Table 1 shows the composition details and the material density determined following the standard test method ASTM 79-08, using anhydrous ethanol. The polymer ratios used were selected in order to obtain materials ranging from dispersed phase morphology to co-continuous morphologies.

Blends and nanocomposites were prepared in a batch mixer (Plasti-Corder, Brabender®) at 240 °C. Prior to the compounding, PA6 was dried in a vacuum oven at 80 °C for 24 h and PP was used as received. PA6 was melted (2 min at 10 RPM),

Table 1 Composition details of blends and nanocomposites based in 100 g of material

Sample	PP g	PA6 g	MWCNTs g	Density ^a (g/cm ³)	PP/PA6/PP-g-GMA	PP g	PA6 g	PP/PP-g-GMA g	MWCNTs g	Density (g/cm ³)
PP	100	0	0	0.910	PP/PP-g-GMA	95.0	0	5	0	0.917
80/20	80	20	0	0.9460	80/20/5	76.0	20	4	0	0.998
60/40	60	40	0	0.9775	60/40/5	57.0	40	3	0	1.012
50/50	50	50	0	0.9987	50/50/5	47.5	50	2.5	0	1.019
40/60	40	60	0	1.029	40/60/5	38.0	60	2	0	1.040
20/80	20	80	0	1.068	20/80/5	19.0	80	1	0	1.073
PA6	0	100	0	1.140	–	–	–	–	–	–
PP/PA6/MWCNTs					PP/PA6/MWCNTs/PP-g-GMA					
PP/MWCNTs	96.0	0.0	4	0.960	PP/PP-g-GMA/MWCNTs	91.20	0.0	4.80	4	0.964
80/20/4	76.8	19.2	4	0.977	80/20/4/5	72.96	19.2	3.84	4	1.016
60/40/4	57.6	38.4	4	0.981	60/40/4/5	54.72	38.4	2.88	4	1.092
50/50/4	48.0	48.0	4	1.023	50/50/4/5	45.5	48.0	2.4	4	1.071
40/60/4	38.4	57.6	4	1.073	40/60/4/5	36.48	57.6	1.92	4	1.092
20/80/4	19.2	76.8	4	1.119	20/80/4/5	18.24	76.8	0.96	4	1.135
PA6/MWCNTs	0.0	96.0	4	1.160	–	–	–	–	–	–

^a The experimental error within 5 %

after which PP was added and mixed (1 min); once the melt was completed, the blend was mixed for 5 min at 50 RPM. In the case of MWCNT nanocomposite preparation, MWCNTs were slowly dispersed in the melt PA6 matrix, and PP was then added to the composite. Compatibilized systems were prepared using a PP/PP-g-GMA blend (95/5 wt%) previously prepared using a Brabender® twin-screw extruder (temperature profile: 180–190–200 °C, speed 50 rpm, 3 mm diameter capillary die). The above PP/PP-g-GMA blend was mixed with the PA6 and MWCNTs in order to obtain the compatibilized PP/PA6 blends and compatibilized PP/PA6/MWCNTs nanocomposites. Thin laminates (120 × 120 mm, 1 mm thick) were compression-molded from the above blends and nanocomposites using a lab press (Carver Inc., USA) at 240 °C and 5 Ton clamping load. After 5 min, the laminates were water-cooled under pressure.

Characterization

Morphology

The morphology of polymer blends, the dispersion and distribution of MWCNTs and the MWCNTs localization in the PP/PA6 blends were determined by means of SEM images from cryogenic fractured surfaces of the samples prepared by compression molding. Images were obtained using a scanning electron microscope, JEOL model JSM-6360 LV. Cryo-fractured samples were etched with nitric acid for 10 min at room temperature in order to remove the PA6 phase in the blends and xylene was used at 120 °C to remove the PP phase. Furthermore, transmission electron microscopy (TEM) was used to observe the dispersion of MWCNTs into PP, PA6 and selected PP/PA6 blends using a TEM FEI Titan 80–300 kV. Samples were cut under cryogenic conditions to obtain thin slices.

Volume resistivity

Volume resistivity of nanocomposites was calculated with the values of the electrical resistance obtained from a high impedance electrometer (Keithley Model 6517B) at room temperature. Rectangular specimens (20 × 3 mm length, and 1 mm thick) were prepared from the plates obtained by compression molding. Silver conductive ink (Ted Pella Inc., USA) was used at the sample edges as electrodes and to reduce contact resistance. The volume resistivity of neat polymer systems was determined on 60 mm square specimens (1 mm average thickness) using a standard metal guard (Keithley 8009). In all cases, five replicates were done.

Mechanical properties

Tensile mechanical properties were evaluated on the compression-molded samples according to ASTM-D638 standard method using a universal testing machine Shimadzu AGS-X and specimens type IV. The total length for the dogbone specimen is 115 mm, gauge length is 33 mm and the narrow section width is 6 mm.

Differential scanning calorimetry (DSC)

Differential scanning calorimetry experiments were performed on selected composites using a Perkin-Elmer DSC-7 calorimeter. The samples were heated from 50 to 250 °C at a heating rate of 10 °C/min under nitrogen atmosphere using aluminum pans. The typical sample weight was about 5 mg. In order to eliminate the thermal history of the samples, two runs were performed and fast cooling was employed between heating runs. The percent of crystallinity X_c was calculated according to Eq. 1.

$$X_c = \frac{\Delta H_c}{\Delta H_c^0 \lambda_c} \times 100 \%, \quad (1)$$

where ΔH_c is the melting enthalpy obtained experimentally, ΔH_c^0 is the melting enthalpy for 100 % crystalline polymer reported in the literature (190.6 and 209 J/g for PA6 and PP, respectively [26]) and λ_c is the polymer content in the blends or nanocomposites.

X-ray diffraction (XRD)

X-ray diffraction patterns were obtained from a D-5000 diffractometer (Siemens) using $\text{CuK}\alpha$ radiation ($\lambda = 1.7818 \text{ \AA}$) at a scan rate of 0.3°/min in the 2θ range 10°–30° operated at 25 mA and 34 kV.

Results and discussion

Morphology

MWCNT dispersion was evaluated in PP, PP/PP-GMA and PA6 nanocomposites to elucidate its effect on the electrical and mechanical properties of nanocomposites and to assess the compatibilizing effect of PP-g-GMA. Figure 1 shows SEM and TEM micrographs of the cryo-fractured surface for PP/MWCNT, (PP/PP-g-GMA)/MWCNT and PA6/MWCNT; the MWCNT content was 4 wt% in all cases. SEM images show nanotubes mainly distributed in the PP matrix as small agglomerations (Fig. 1a). The poor nanotube dispersion into PP could be explained by several factors such as, the low affinity of MWCNTs to PP, the high attractive forces (van der Waals, electrostatic attraction, etc.) among nanotubes, and polymer viscosity [20]. TEM images (Fig. 1b) for the PP system show clear fields and sections with high content of nanotubes; these results confirmed the observations by SEM discussed above. Other studies have reported an inadequate dispersion of MWCNTs into PP matrix, especially when the MWCNTs are added in high concentrations [40–42]. Figure 1c, d shows the MWCNT dispersion into PP/PP-g-GMA composite; SEM and TEM images suggest better nanotube dispersion compared with PP/MWCNT composites. The use of compatibilizers, such as polypropylene grafted Maleic Anhydride (PP-g-MA), has been reported to produce a similar effect [42]. In

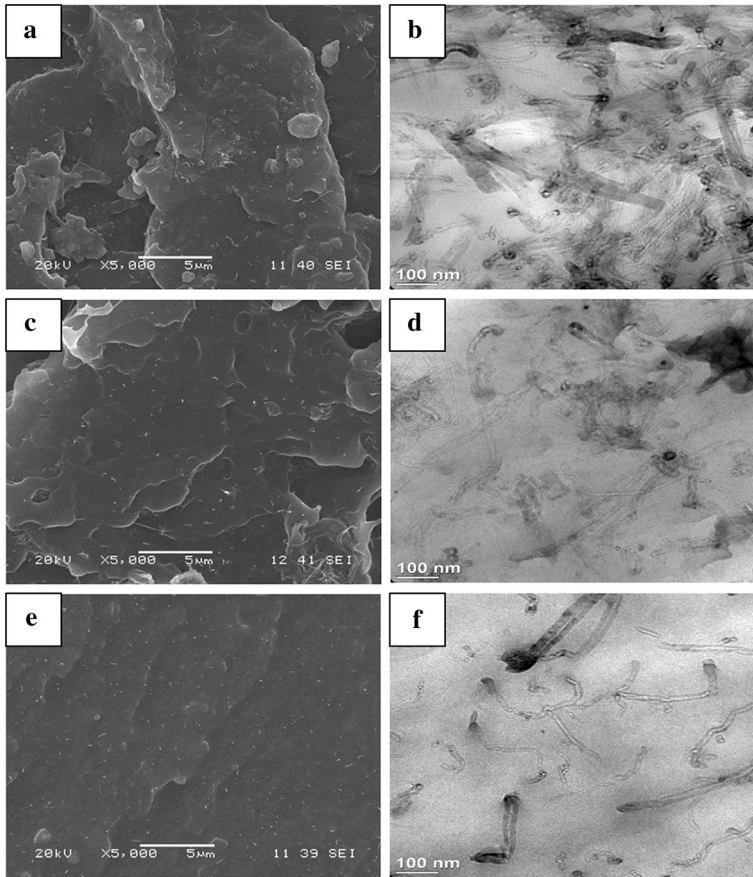


Fig. 1 SEM (*left*) and TEM (*right*) images of PP/MWCNT (**a, b**), PP/PP-g-GMA/MWCNT (**c, d**) and PA6/MWCNT (**e, f**) nanocomposites

this case, epoxy groups from PP-g-GMA are expected to generate interactions with polar groups of nanotubes, resulting in improved nanotube dispersion within the PP matrix. Indeed, Chen et al. [43] reported a possible nucleophilic substitution reaction between the epoxy group from PP-g-GMA and carboxyl groups of carbon nanotubes in melt compounded PP-g-GMA/CNT composites. Figure 1e, f corresponds to SEM and TEM micrographs for PA6/MWCNT nanocomposites, MWCNTs are better dispersed in comparison with the PP systems.

The differences in the degree of nanotube dispersion could be attributed to several factors such as, interactions among the polar groups from PA6 and the functional groups located on the nanotube surface, and the low melt viscosity and high polarity of PA6 matrix in comparison with PP resin. Similar results have been reported in the literature; for instance, Moud et al. [20] studied the dispersion of MWCNTs in PP and PA6 composites by TEM. MWCNTs are better dispersed in PA6 due to the existence of interactions between the nanotubes and polar PA6

matrix. In contrast, the dispersion of MWCNTs into the PP matrix was poor, nanotubes were observed as large agglomerations because of a weak interaction between the nanotubes and the nonpolar PP matrix.

The effect of the composition of polymer blend, MWCNT load (4 wt%), and PP-GMA on the morphology of composites was determined by SEM. Selective degradation was used to remove a polymer phase to reveal morphological details; nitric acid treatment was used to remove the PA6 phase when polyamide was at low content (i.e. 20–60 wt%), in consequence the remaining material in the SEM pictures is the PP phase and the empty domains correspond to PA6 phase. At high PA6 content (80 wt%), the samples were treated with hot Xylene to dissolve the PP phase and the remaining material is PA6 phase.

Figure 2a shows the cryo-fractured surface of the blend of PP/PA6 (80/20), the typical dispersed phase morphology was observed. The PA6 phase was dispersed mainly as spherical domains in different diameters. This morphology indicates the lack of affinity between the phases; the main reasons leading to the formation of incompatible systems are the absence of specific interactions between the blend components, the differences between the viscosities of each polymer and structural differences [28]. A totally different morphology was observed for the PP/PA6/PP-g-GMA blend (80/20/5) (Fig. 2b); in this case, the size of domains was reduced and

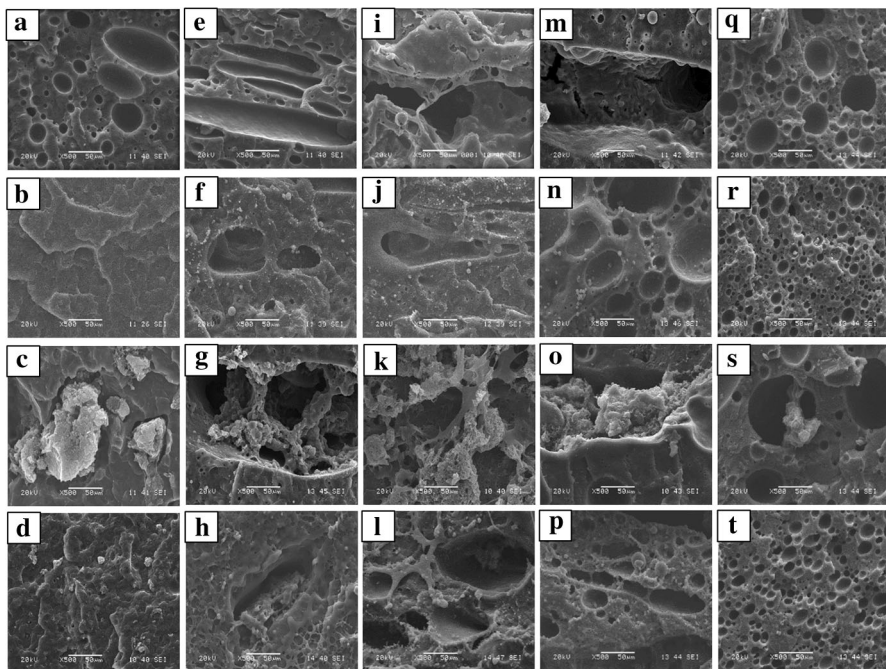


Fig. 2 SEM images of uncompatibilized and compatibilized systems (PP/PA6 blends and PP/PA6/MWCNT nanocomposites) with different polymer compositions: **a** 80/20, **b** 80/20/5, **c** 80/20/4, **d** 80/20/4/5, **e** 60/40, **f** 60/40/5, **g** 60/40/4, **h** 60/40/4/5, **i** 50/50, **j** 50/50/5, **k** 50/50/4, **l** 50/50/4/5, **m** 40/60, **n** 40/60/5, **o** 40/60/4, **p** 40/60/4/5, **q** 20/80, **r** 20/80/5, **s** 20/80/4, **t** 20/80/4/5

these were homogeneously distributed. PP-g-GMA leads to an increase in the interfacial adhesion between PP and PA6. After the addition of MWCNTs to the PP/PA6 (80/20) blend (Fig. 2c), the PA6 phase still presents the same morphology (dispersed phase), but the dispersed phase changes from a spherical shape to an irregular shape. The change in the shape of the PA6 phase could be related to the increased viscosity of the PA6 phase due to the preferential localization of the MWCNTs in this phase as reported by Zhang et al. [15]. As the compatibilizer was added to the PP/PA6/MWCNT 80/20/4 nanocomposite, the size of dispersed phase morphology was reduced (Fig. 2d), this would suggest that PP-g-GMA promotes better interfacial adhesion between the nanocomposite components.

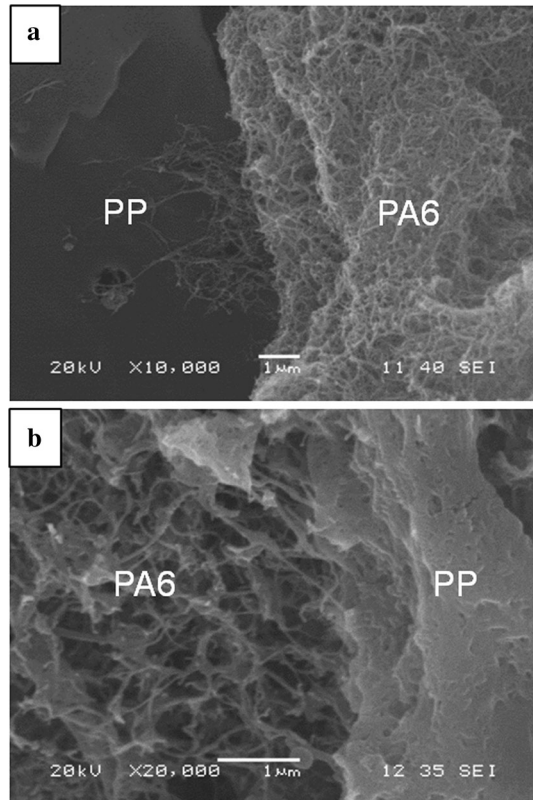
As the PA6 concentration increased (PP/PA6 60/40), PA6 domains became larger (Fig. 2e) exhibiting an elongated shape which could be due to coalescence of the PA6 domains due to increased PA6 content. PP-g-GMA prevented the coalescence of PA6 phase avoiding the formation of elongated domains (Fig. 2f). Adding MWCNTs to the PP/PA6 (60/40) blend provoked an increase in the size of the PA6 phase due to the selective localization of nanotubes in the PA6 phase (Fig. 2g). With the addition of PP-g-GMA, the size of the dispersed phase decreased slightly and nanotubes were selectively located in the PA6 phase (Fig. 2h).

Furthermore, when the PA6 concentration was increased (i.e. 50/50 and 40/60) in the blends, the resulting materials exhibited a continuous phase morphology (Fig. 2i, m, respectively). On the other hand, the PA6 phase changed from a continuous phase to dispersed phase in the compatibilized blends (Fig. 2j, n, respectively). The morphology of uncompatibilized blends (50/50 and 40/60) remained unchanged even when MWCNTs were added (Fig. 2k, o). The addition of PP-g-GMA provoked an important decrease in the size of the domains of polymer blends (Fig. 2l, p). As the PA6 content increased (i.e. 80 wt%), PP became the dispersed phase and was distributed into the PA6 matrix as spherical domains (Fig. 2q). PP-g-GMA reduced significantly the domain size of the PP phase (Fig. 2r).

Figure 2s shows that the addition of MWCNTs to the blend increased the PA6 domains, nanotubes were not seen in the PP dispersed phase which confirmed the selective location of nanotubes in the PA6 phase. Finally, Fig. 2t shows the morphology for the compatibilized 20/80/4 nanocomposite in which a reduction of dispersed PP domains can be observed. SEM micrographs suggest that MWCNTs are preferably located in the PA6 phase regardless of the morphology of the blends (continuous or dispersed phase), and MWCNTs are dispersed in the PA6 phase in the form of agglomerates due to the increase in the effective MWCNT concentration.

Figure 3 shows a close-up of the morphologies for uncompatibilized PP/PA6/MWCNT nanocomposites (80/20/4 and 40/60/4, respectively) after removing the PA6 phase. Figure 3a shows some nanotubes crossing from the PA6 phase to PP phase in the 80/20/4 nanocomposite, possibly because the length of the nanotubes is greater than the PA6 domains [15, 25]. Figure 3b shows nanotubes contained in the PP phase because the PA6 domains are larger in comparison with the PA6 domain from the 80/20/4 nanocomposite.

Fig. 3 SEM images of PP/PA6/MWCNT nanocomposites after PA6-phase extraction: **a** 80/20/4, and **b** 40/60/4



The dispersion and selective localization of the carbon nanotubes in the PA6 phase were also studied by TEM in selected nanocomposites with different morphologies (dispersed and continuous morphology of PP/PA6). Figure 4a, b shows TEM images at different magnifications corresponding to PP/PA6/MWCNT (80/20/4) nanocomposite; the nanotubes are selectively located in the minor PA6 phase. On the other hand, Fig. 4c, d shows the phase boundary of the co-continuous morphology for PP/PA6/MWCNT (40/60/4) nanocomposite, CNTs are selectively localized within the PA6 phase and homogenously dispersed.

Volume resistivity

Figure 5 shows the volume resistivity results for neat PP, PA6 and PP/PP-*g*-GMA resins, and the corresponding MWCNT nanocomposites (4 wt%). The results show that PA6 and PP polymers exhibit high volume resistivity (10^{15} and 10^{13} Ω cm, respectively), and the addition of PP-*g*-GMA did not modify the volume resistivity of the blend. The high resistivity values of PP and PA6 decreased with the addition of MWCNTs in a different fashion.

The results also show that PP/MWCNT nanocomposites exhibit lower volume resistivity compared to PA6/MWCNT nanocomposites. The volume resistivity for

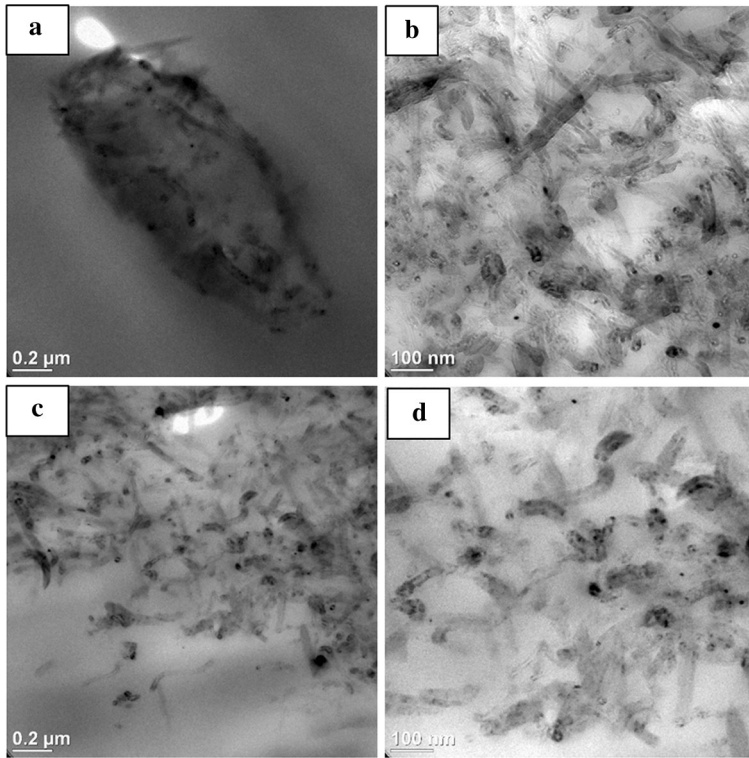


Fig. 4 TEM images of PP/PA6/MWCNT nanocomposites at different magnifications: 80/20/4 (**a**, **b**) and 40/60/4 (**c**, **d**)

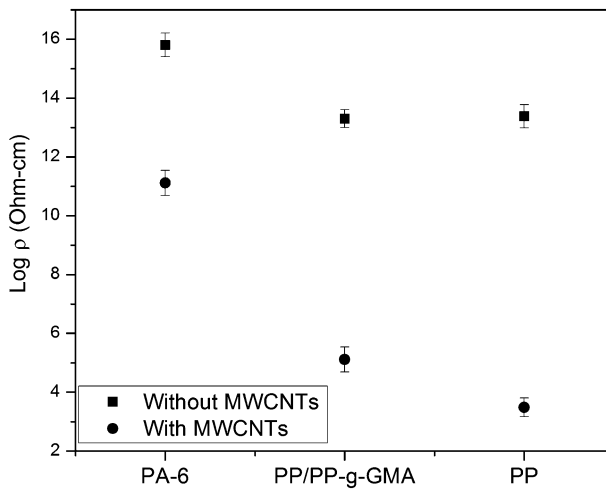


Fig. 5 Volume resistivity of PA6, PP/PP-g-GMA and PP and their corresponding MWCNT nanocomposites (4 wt%)

PA6/MWCNT nanocomposites decreased four orders of magnitude ($10^{11} \Omega \text{ cm}$) compared to pure PA6 ($10^{15} \Omega \text{ cm}$); similar results were obtained by Meincke et al. [16]. The high volume resistivity of PA6/MWCNT nanocomposites is attributed to the homogeneous dispersion and the presence of fewer agglomerates of carbon nanotubes within the PA6 matrix, as observed in the SEM and TEM images. Krause et al. [44] reported a reduction in the number and size of MWCNT agglomerates in PA6/MWCNT nanocomposites leading to higher electrical volume resistivity, in comparison with PA6/MWCNT nanocomposites with larger agglomerates of nanotubes. The homogeneous dispersion of carbon nanotubes could reduce the probability of contact among nanotubes, preventing the formation of conductive networks in the PA6 matrix [20]. On the other hand, several studies have suggested that individual or agglomerates of MWCNTs could be coated with a thin polymer layer due to the high interaction between the polar PA6 and the polar moieties from the MWCNT surface [44–47]. Thus, as the contact resistance between the carbon nanotubes increases, the formation of conductive network becomes more difficult.

The volume resistivity of PP/MWCNT ($10^3 \Omega \text{ cm}$) nanocomposites was 10 orders of magnitude lower in comparison with neat PP matrix ($10^{13} \Omega \text{ cm}$). The poor dispersion of MWCNT generated agglomerates within the PP phase, leading to the formation of interconnected CNT aggregates [20]. Additionally, it is possible that the contact resistance between the nanotubes was lower due to the reduced possibility that carbon nanotubes were coated with a PP layer as a result of the low affinity between PP and nanotubes. Figure 6 shows that the volume resistivity of PP/PP-g-GMA is comparable to that of the PP. However, the volume resistivity for PP/PP-g-GMA/MWCNT nanocomposite was two orders of magnitude ($10^5 \Omega \text{ cm}$) higher than the PP/MWCNT nanocomposite ($10^3 \Omega \text{ cm}$). The compatibilizer improves the dispersion of nanotubes in the PP matrix (as observed by SEM)

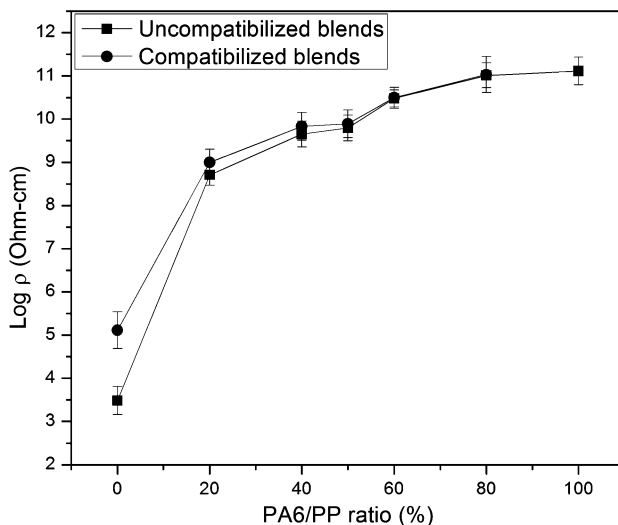


Fig. 6 Volume resistivity for uncompatibilized and compatibilized PP/PA6/MWCNT nanocomposites

decreasing the number and size of the nanotube agglomerates within the PP matrix and, as a result, the volume resistivity increases. This result is consistent with other studies on PP/MWCNT composites using PP-*g*-MA as compatibilizer [42, 48].

Figure 6 shows volume resistivity of PP/PA6/MWCNT (uncompatibilized and compatibilized) nanocomposites as a function of polymer ratio. The results show that volume resistivity of composites depended on the polymer content: nanocomposites with high content in PA6 exhibited higher volumetric resistivity; the lowest volume resistivity value was recorded at the lowest PA6 content (80/20/4). This behavior could be explained in terms of the relative concentration of nanotubes in the PA6 phase and the morphology. In the former case, as the content of PA6 in the blend decreases the relative concentration of nanotubes in the PA6 phase increases. The morphology of the dispersed PA6 phase in the 80/20 blend could favor the contact between the MWCNTs, forming conductive networks in the blend. These observations are consistent with other studies carried out on PP/PA6 blends [3]. Furthermore, nanotubes leaving the PA6 phase (Fig. 3a) could contribute to the generation of a conductive percolation mesh.

Interestingly, the volume resistivity values for compatibilized PP/PA6/MWCNT nanocomposites are comparable to those of the uncompatibilized nanocomposites; this fact suggests that the concentration of PP-*g*-GMA used was probably not sufficient to modify the preferential localization and effective concentration of MWCNTs in the PA6 phase.

Tensile mechanical properties

The mechanical properties for PP, PA6 and their uncompatibilized and compatibilized blends are shown in Fig. 7; PA6 generally exhibits better mechanical properties than PP. Tensile modulus and tensile strength for PP/PA6 blends are lower than those of pure polymers. The immiscibility between PP and PA6 generates a two-phase system with a weak interface and poor mechanical properties [16, 18, 26]. However, adding PP-*g*-GMA to the PP/PA6 blends improves the mechanical properties significantly. The ultimate strain for the PP/PA6 blends is directly related to the proportion of the PA6 in the blend; as PA6 content increases, the deformation of the blend also increases. The reduced value of ultimate strain for pure PP could be caused by the processing conditions employed [49].

The mechanical properties of PP/PA6/MWCNT nanocomposites are summarized in Fig. 8. The results show that the addition of MWCNTs produced an increase in the tensile modulus due to the high modulus of MWCNTs and their selective location in the PA6 phase. Similarly, the results show that ultimate strain and tensile strength tend to be lower in comparison with PP/PA6 blends. The decrease in the mechanical properties may be attributed to several factors such as, the relatively high concentration of MWCNTs employed (4 wt%), the selective localization of MWCNTs within the PA6 phase and the quality of the interface between the polymers.

In the first case, Wang et al. [50] reported that the mechanical properties of polyimide (PI)/Amino-functionalized carbon nanotube (CNT-ODA) composites were enhanced with a CNT load up to 3 wt%; however, at higher CNT content

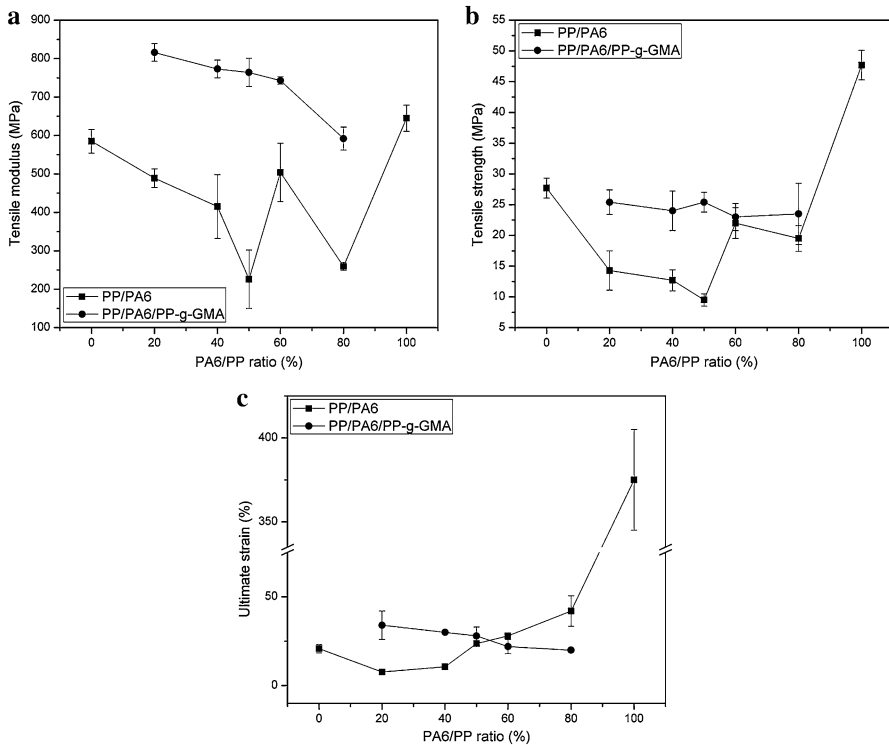


Fig. 7 Mechanical properties for PP/PA6 and PP/PA6/PP-g-GMA blends; **a** tensile modulus, **b** tensile strength and **c** ultimate strain

(5 wt%), the tensile strength, Young's modulus, and elongation at break decreased; similar behavior has been observed in other conductive nanocomposites [51]. A number of investigations have also reported poor mechanical properties for MWCNT nanocomposites at higher MWCNT content, above 2 wt%, due to the formation of MWCNT agglomerates [4, 26].

The selective localization of MWCNTs within the PA6 phase favors the formation of MWCNT agglomerates which act as stress concentrators, causing a decrease in the mechanical properties of the composite [26]. Finally, the mechanical properties of immiscible blends depend on the quality of the interface between the polymers; carbon nanotubes are not effective in improving interfacial adhesion in immiscible blends [16, 26, 52].

The effect of PP-g-GMA on the mechanical properties of PP/PA6/MWCNT nanocomposites was investigated. In general, the mechanical properties of PP/MWCNT and PP/PA6/MWCNT nanocomposites were improved by adding 5 wt% of PP-g-GMA to the PP phase (Fig. 8). PP/PP-g-GMA/MWCNT nanocomposites have better mechanical properties in comparison with PP/MWCNT nanocomposites, which can be attributed to a better dispersion and the physicochemical interactions between nanotubes and PP/PP-g-GMA blend. The mechanical properties of compatibilized PP/PA6/MWCNT composites were also improved; PP-g-

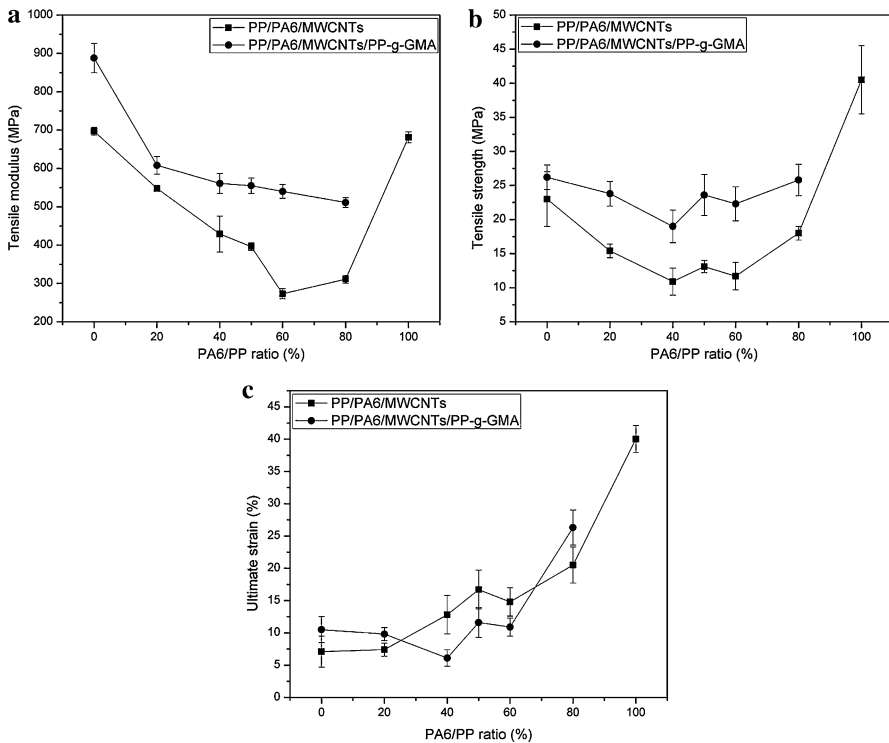


Fig. 8 Mechanical properties for PP/PA6/MWCNT and PP/PA6/PP-g-GMA/MWCNT nanocomposites; **a** tensile modulus, **b** tensile strength and **c** ultimate strain

GMA promotes the interactions between PP and PA6 phases at the interface, improving the interfacial adhesion between the polymers, as observed in the SEM micrographs. The effect of a compatibilizer on the mechanical properties of PP/PA6 blends has been reported for systems containing maleic anhydride copolymers; the structural stability and morphology of the blends were improved due to the in situ reaction among the anhydride groups and the amino end-groups of PA6 [53].

In the case of PP/PA6/PP-g-GMA blends, the epoxy groups from PP-g-GMA could react with either carboxyl or amino end-groups of PA6 resin to form compatibilized PP-g-PA6 blends, as described by Tedesco et al. [39]. In this sense, the simultaneous addition of MWCNTs and PP-g-GMA to PP/PA6 blends could produce PP-g-PA6/MWCNT nanocomposites where MWCNTs are preferentially located in the PA6 phase (due to high affinity and interaction of PA6 and MWCNTs); the above is in agreement with similar studies related to PP/PA6/MWCNT nanocomposites compatibilized with polypropylene grafted maleic anhydride [33]. The results suggest that PP-g-GMA has potential to be used as compatibilizer for this polymer system.

Additional characterization was carried out on the composites (PP/PA6, 80/20) exhibiting both good electrical and mechanical properties in order to investigate the

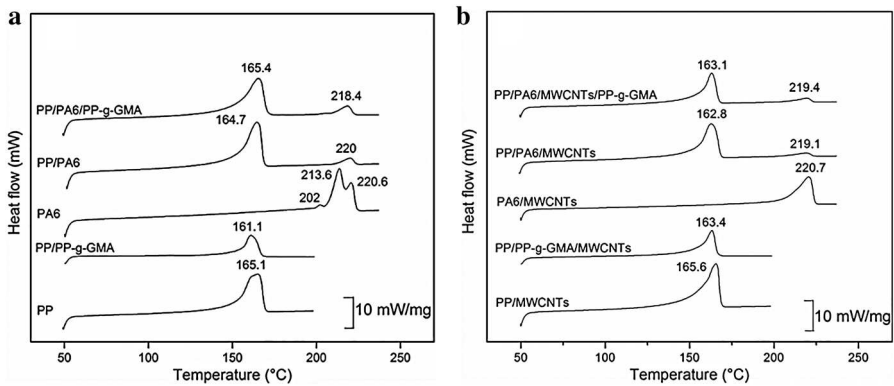


Fig. 9 DSC thermograms for **a** PP, PP-g-GMA, PA6, PP/PA6 blends, and **b** corresponding MWCNT nanocomposites

effect of MWCNTs and compatibilizer on the polymer crystallinity. Figure 9a shows the DSC thermograms corresponding to the second heating run for neat polymers (PP, PA6 and PP-g-GMA) and uncompatibilized and compatibilized PP/PA6 (80/20) blends, and Fig. 9b shows the thermograms for the corresponding MWCNT nanocomposites. Table 2 summarizes thermal parameters of interest. According to the results, PP shows one melting point at 165.1 °C in contrast with PA6 which shows three melting points (202, 213.6 and 220.6 °C) corresponding to different crystallographic forms. The melting point at 220.6 °C is associated with α crystalline form, the second one at 213.6 °C corresponds to the melting of the γ crystalline form, in agreement with [26, 54]. The third peak at ~ 202 °C is attributed to a less perfect χ -crystallite form [55]. Regarding the uncompatibilized

Table 2 Melting temperatures and percentages of crystallinity

Sample	T_m °C		X_c (%)	
	PP	PA6	PP	PA6
PP	165.1	–	39.1	–
PP/MWCNTs	165.6	–	41.6	–
PA6	–	202, 213.6, 220.6	–	22.9
PA6/MWCNTs	–	220.6	–	34.1
PP/PP-g-GMA	161.1	–	43.1	–
PP/PP-g-GMA/MWCNTs	163.4	–	58.2	–
PP/PA6	164.7	220	41.3	14.2
PP/PA6/MWCNTs	162.8	219.1	42.0	16.6
PP/PA6/PP-g-GMA	165.4	218.4	40.8	24.5
PP/PA6/MWCNTs/PP-g-GMA	163.1	219.4	42.4	18.7

blend, the thermogram shows the melting point for PP at 164.7 °C and the melting point for PA6 at 220 °C corresponding to the α crystalline form; similar results have been reported in the literature [30]. It has been observed recently in PP/PA6 blends that one blend component could act as a nucleating agent [20].

MWCNTs increase the melting point and crystallinity for both PA6 and PP (Fig. 9b); suggesting that MWCNTs could act as a nucleating agent [56]. Additionally, nanotubes promote the formation of the more stable α crystal form for PA6 [26]. In contrast, adding MWCNTs to PP/PA6 blend decreases the melting point for both PP and PA6, but the crystallinity increases; this behavior would suggest the formation of crystals with a lower degree of perfection due to the restriction of polymer chain mobility. Further increase in the crystallinity for PA6 could be explained by the preferential localization of the MWCNTs in the PA6 phase. The incorporation of PP-*g*-GMA to the PP/PA6 blend produces a significant increase in the crystallinity of PA6; however, the addition of MWCNTs to the system decreases the crystallinity of PA6. The decrease in PA6 crystallinity could be related to the high content of MWCNTs as explained above.

The XRD patterns for PP, PA6 and PP/PA6 blends are shown in Fig. 10a. Neat PA6 shows two distinct peaks at $2\theta = 20.2^\circ$ and 23.9° corresponding to the reflection planes (200), (002) and (202) associated with the α -crystalline form. Neat PP shows mainly the α -crystalline phase [peaks at $2\theta = 14.1^\circ, 16.9^\circ, 18.5^\circ, 21.1^\circ, 21.7^\circ, 25.4^\circ$, corresponding to the reflection planes (110), (040), (130), (111), (041) and (060)] [57]. The XRD pattern for PP/PA6 (80/20) shows the corresponding peaks of PP and PA6, the intensity of the peaks depends on the polymer concentration. The addition of PP-*g*-GMA favors the formation of the α -crystalline form of PA6. The presence of MWCNTs does not affect the crystal structure of PP or PA6 (Fig. 10b).

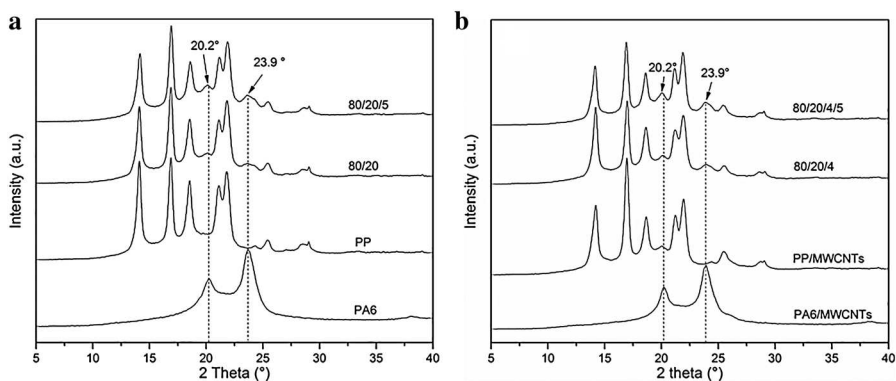


Fig. 10 X-ray diffraction patterns of **a** PP, PP-*g*-GMA, PA6, PP/PA6 blends and **b** corresponding MWCNTs nanocomposites

Conclusions

The performance of PP-*g*-GMA as compatibilizer for PP/PA6 blends and their MWCNT nanocomposites was evaluated based on the morphology, electrical and mechanical properties. SEM and TEM images indicate that nanotubes were better dispersed in PA6. Despite the better dispersion of carbon nanotubes in PA6, PA6/MWCNT nanocomposites exhibited higher volume resistivity than PP/MWCNTs. SEM images confirmed that the addition of PP-*g*-GMA to PP improved the dispersion of MWCNTs, producing an increase in the volume resistivity of nanocomposites. The addition of PP-*g*-GMA to PP/PA6 blends and to the PP/PA6/MWCNT nanocomposites was found to decrease the size of PA6 domains regardless of the morphology of composites (dispersed or continuous-phase).

Volume resistivity of PA6/MWCNT nanocomposites decreased with the addition of PP due to the increase in MWCNT concentration in the PA6 phase. The volume resistivity for compatibilized PP/PA6/MWCNT nanocomposites did not vary significantly with respect to the uncompatibilized nanocomposites. The above suggests that the PP-*g*-GMA concentration used was insufficient to modify the preferential localization of MWCNTs for the PA6 phase.

The addition of MWCNTs increased tensile modulus and decreased both strength and ultimate strain for the PP, PA6 and PP/PA6 nanocomposites. PP-*g*-GMA promoted MWCNT dispersion within PP, resulting in improved mechanical properties. The incorporation of PP-*g*-GMA to PP/PA6 blends and PP/PA6/MWCNT nanocomposites produced an increase in the mechanical properties while preserving the electrical properties.

Additional characterization was carried out on the composites (PP/PA6, 80/20), exhibiting interesting mechanical and electrical properties. The presence of PP, PP-*g*-GMA or MWCNT promoted the formation of α crystalline form for PA6 phase. The degree of crystallinity of PP tended to increase when it was combined with MWCNT, PP-*g*-GMA or PA6; in contrast, the degree of crystallinity of PA6 tended to decrease. XRD results suggest that the transition of α crystalline form of PA6 to a less stable form is due to the heating-cooling process during DSC analysis.

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